

Verification of the Beer-Lambert Law via Laser Attenuation in Dye Solutions

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This experiment empirically validates the Beer-Lambert Law, which predicts that the optical absorbance of a solution scales linearly with both the concentration of the absorbing species and the optical path length. Using a 660 nm laser with a 12.4 mW power output and a blue dye solution, we measured optical power attenuation across path lengths ranging from 1 to 10 cm and concentrations spanning a factor of 16 (0.0625 to 1.0 arbitrary units). To isolate absorption from reflection losses at the air-glass and glass-liquid interfaces, we employed a pairwise measurement approach, defining transmittance as the ratio of power transmitted through the dye solution to that through pure water in the same optical cell. Analysis of the data in logarithmic space using Orthogonal Distance Regression (ODR) yielded a weighted mean slope of $1.003 \pm 0.004 U^{-1}$ for absorbance versus concentration, and weighted least-squares regression yielded a slope of $1.053 \pm 0.043 \text{ cm}^{-1}$ for absorbance versus path length. Both results are consistent with the theoretical prediction of unity to within 1.5σ . Furthermore, all fitted intercepts were statistically consistent with zero, confirming that attenuation vanishes in the limits of zero path length or concentration. The absorptivity constant for the system was determined to be $\varepsilon = 0.1154 \pm 0.0025 [\text{U}\cdot\text{cm}]^{-1}$. These findings confirm the Beer-Lambert Law ($A = \varepsilon LC$) as a robust model for optical absorption in this regime.

I. INTRODUCTION

The interaction between light and matter is central to a wide range of physical, chemical, and biological processes. One of the simplest and most powerful models describing this interaction is the Beer-Lambert law, which predicts that the intensity of a light beam decreases exponentially as it passes through an absorbing medium. From spectroscopy and optical sensing to chemical analysis, this relationship underpins countless experimental techniques.

Although the Beer-Lambert law is well established theoretically, current work aimed at experimental validation of the law remains essential. Our experiment examines laser attenuation through dye solutions to verify the law's prediction that transmitted power depends systematically on both absorber concentration and optical path length. Five dye concentrations and six cell lengths were used to probe this dependence and to evaluate how closely real measurements adhere to the ideal model.

A. Physical Expectations

Before designing the measurement protocol, we establish physical expectations for the system. As the path length L through the dye solution gets smaller, the laser encounters less absorbing material, and we expect the transmitted power to approach the incident power—that is, attenuation vanishes as $L \rightarrow 0$. Similarly, as the dye concentration C approaches zero, the solution becomes transparent to the laser, and attenuation again vanishes. These limiting cases suggest that both L and C play roles in determining the degree of optical absorption, with attenuation increasing as either parameter increases.

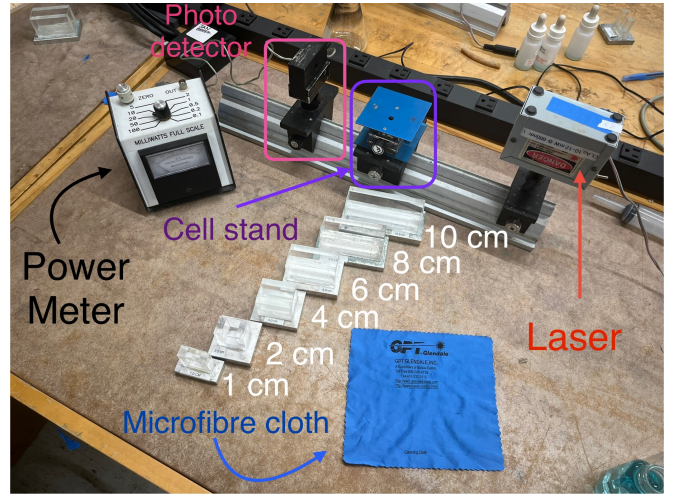


FIG. 1. Schematic of the experimental setup showing the laser, cells stand, photo-detector, power meter, the cells (and their lengths), and the microfibre cloth.

B. The Challenge of Interface Reflections

A significant experimental complication arises from reflections at the multiple interfaces in the optical path. Each air-glass and glass-liquid boundary reflects a fraction of the incident light. For an optical cell with outer surfaces exposed to air and inner surfaces in contact with the liquid, there exist two distinct transmission coefficients: T_1 for the air-glass interfaces and T_2 for the glass-liquid interfaces. These coefficients represent the fraction of incident power transmitted through each boundary (e.g., if 4% is reflected, $T = 0.96$).

Importantly, we assume that the transmission coeffi-

cients at the internal glass-liquid boundaries are independent of dye concentration; they depend only on the refractive indices of the glass and the solvent. If the dye solution caused no absorption whatsoever, the transmitted power would be $P_{\text{trans}} = T_1^2 T_2^2 P_0$, accounting for the four interfaces (entrance and exit on both outer and inner surfaces). Attempting to measure and correct for each reflected beam individually is experimentally challenging and time-consuming. Instead, we require a measurement strategy that inherently eliminates the contribution of these reflection losses.

C. Normalization via Transmittance

The solution to this experimental challenge is to perform a *pairwise measurement*. For each optical cell of length L , we measure two quantities:

- P_{water} : the transmitted power through the cell filled with pure solvent (water)
- P_{dye} : the transmitted power through the same cell filled with dye solution at concentration C

Both measurements experience identical reflection losses since the refractive index of the dilute dye solution is effectively that of water. By forming the ratio

$$T = \frac{P_{\text{dye}}}{P_{\text{water}}}, \quad (1)$$

we define a *normalized transmittance* that isolates the attenuation due solely to the dye molecules. The reflection coefficients T_1 and T_2 cancel in this ratio, leaving only the exponential attenuation caused by absorption within the solution. This dimensionless quantity T ranges from 0 (complete absorption) to 1 (no dye-induced attenuation).

D. The Beer-Lambert Law

The Beer-Lambert Law[1] is the empirical relationship that quantizes the physical reasoning outlined above. Validated across countless chemical systems, it states that the transmittance decays exponentially:

$$T = e^{-\varepsilon LC}, \quad (2)$$

where ε is the absorptivity coefficient, a molecular property characterizing the strength of absorption. Equivalently, defining the *absorbance* $A = -\ln(T)$, the Beer-Lambert Law becomes:

$$A = \varepsilon LC. \quad (3)$$

The logarithmic transformation converts the exponential decay in transmittance into a linear dependence, making ε directly accessible through the slope of A versus

L (at fixed C) or A versus C (at fixed L). The objective of this experiment is to empirically verify this functional form and extract the extinction coefficient from precision optical measurements.

II. METHODS

The experiment reduced to measuring how the optical power of a laser changed as a function of the dye concentration and the distance the laser traveled through the solution.

A. Sample Preparation

We utilized five distinct dye concentrations. The solutions were prepared via serial dilution: 30 units of the previous stock solution were mixed with 30 units of DI water. To ensure homogeneity, the liquid was poured back and forth between the graduated cylinder and the vial multiple times.

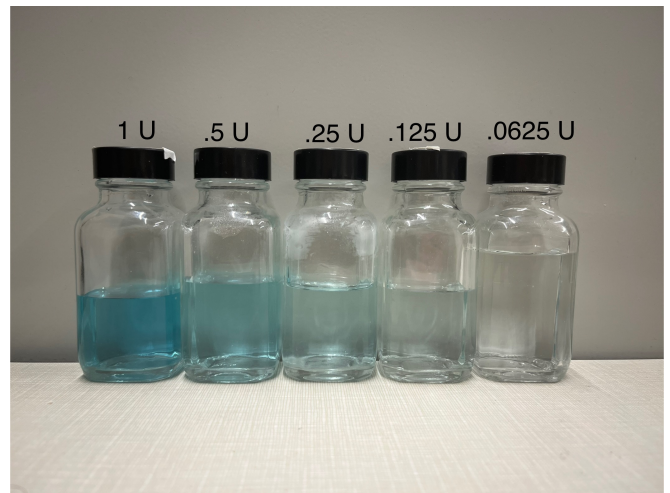


FIG. 2. The five dye solutions prepared via serial dilution. The .0625 U solution is almost indistinguishable to water by eye.

B. Measurement Procedure

Measurements were taken by placing a cell (containing either DI water or dye) between the laser and the power meter. The meter displays optical power as a fraction of a selected full-scale range (e.g., 20 mW or 10 mW), where the maximum reading corresponds to full needle deflection. Because the reading uncertainty is fixed relative to the scale markings (typically ± 0.5 divisions), relative error is minimized by choosing the smallest scale that does not saturate. For example, a beam with 0.97 mW of power should be measured on the 1 mW scale rather

than the 2 mW scale to maximize resolution and read-out precision. Following this protocol, we dynamically adjusted the scale for each trial to ensure the needle deflection was as large as possible. We utilized variable path lengths: 10, 8, 6, 4, 2, and 1 cm.

To mitigate systematic errors, such as instrument drift and non-linearity between different scales—we employed a the following pairwise measurement approach. For every data point:

1. The power meter was zeroed on the 10-scale (or the scale to be used for dye).
2. The cell was filled with DI water, and P_W was measured (on the 20-scale).
3. The cell was emptied, dried with a paper towel, and polished with a microfiber cloth to remove spots that could scatter light.
4. The cell was filled with dye solution, and P_D was measured immediately (either on the 20 or 10 scale).

This ensured that the "zero" reference and the ambient lighting conditions were consistent for every pair of P_W and P_D . Care was taken to eliminate bubbles in the chamber to prevent diffraction losses.

C. Comparison to Prior Methodology

It is worth noting that a previous iteration of this experiment failed due to poor methodology. In that attempt, all water measurements were taken first, and dye measurements were taken much later. This introduced significant drift errors. Furthermore, the scales were not consistently zeroed, and the lowest concentration (0.0625) was too dilute to distinguish from water. Results from this flawed attempt are briefly discussed in the Appendix. The current pairwise method proved vastly superior.

D. Error Propagation and Uncertainty Analysis

Error analysis was conducted to account for uncertainties arising from solution preparation, instrument precision, and data acquisition. We identify three primary sources of uncertainty: concentration C , measured optical power P , and path length L . The path lengths are determined by manufactured optical cells with precision significantly exceeding all other error sources in this experiment, and are therefore treated as exact values with negligible uncertainty.

1. Concentration Uncertainty

The dye concentrations were prepared via serial dilution, a process in which errors accumulate with each suc-

cessive dilution step. For each dilution, we measured two independent volumes using a graduated cylinder with an estimated uncertainty of $\delta V = 0.5$ mL:

$$V_{\text{dye}} = 30 \text{ mL}, \quad V_{\text{water}} = 30 \text{ mL}. \quad (4)$$

The dilution factor is given by

$$D = \frac{V_{\text{dye}}}{V_{\text{dye}} + V_{\text{water}}} = \frac{30}{60} = 0.5. \quad (5)$$

To propagate the volume uncertainties through to the dilution factor, we apply the standard error propagation formula:

$$\delta D = \sqrt{\left(\frac{\partial D}{\partial V_{\text{dye}}} \delta V\right)^2 + \left(\frac{\partial D}{\partial V_{\text{water}}} \delta V\right)^2}. \quad (6)$$

Computing the partial derivatives:

$$\frac{\partial D}{\partial V_{\text{dye}}} = \frac{V_{\text{water}}}{(V_{\text{dye}} + V_{\text{water}})^2} = \frac{30}{3600} \approx 0.0083 \text{ mL}^{-1}, \quad (7)$$

$$\frac{\partial D}{\partial V_{\text{water}}} = -\frac{V_{\text{dye}}}{(V_{\text{dye}} + V_{\text{water}})^2} = -\frac{30}{3600} \approx -0.0083 \text{ mL}^{-1}, \quad (8)$$

we find

$$\delta D = \sqrt{(0.0083 \times 0.5)^2 + (-0.0083 \times 0.5)^2} \approx 0.0059, \quad (9)$$

yielding a relative uncertainty of

$$\frac{\delta D}{D} = \frac{0.0059}{0.5} \approx 0.012 \approx 1.2\%. \quad (10)$$

Thus, each dilution step introduces approximately a 1.2% relative uncertainty. Since each new concentration is formed from the previous one according to $C_{\text{new}} = C_{\text{old}} \times D$, the uncertainties accumulate in quadrature over N dilution steps:

$$\frac{\delta C}{C} = \sqrt{N} \times 0.012. \quad (11)$$

By definition, the stock solution concentration is assigned the normalized value $C = 1.0$ unit with zero uncertainty, serving as the reference against which all diluted concentrations are compared. For the most dilute concentration in this experiment ($C = 0.0625$, corresponding to $N = 4$ dilutions), the relative uncertainty reaches approximately 2.4%.

2. Power Measurement Uncertainty

The optical power meter employs an analog needle display with a selectable full-scale range (e.g., 2 mW, 10 mW, 20 mW). Each power measurement is recorded

as a pair (S, R) , where S is the selected full-scale setting and R is the needle reading on a 0–100 scale. The physical optical power is then calculated as

$$P = S \times \frac{R}{100}. \quad (12)$$

The uncertainty in P arises from two independent sources, which combine in quadrature.

a. Reading Error. The analog display is graduated with tick marks every 2 units. We conservatively estimate the needle position can be read to within ± 1.0 unit. This reading uncertainty translates to a power uncertainty that scales with the full-scale setting:

$$\delta P_{\text{read}} = S \times \frac{1.0}{100}. \quad (13)$$

b. Linearity and Accuracy Error. According to the manufacturer specifications, the meter is linear in incident optical power to within 1%, and the relative accuracy between different scale settings is also approximately 1%, which we model as a fractional uncertainty:

$$\delta P_{\text{acc}} = 0.01 \times P. \quad (14)$$

It's worth noting that the water baseline measurement P_{water} and the dye solution measurement P_{dye} were often acquired on different full-scale settings to optimize needle deflection and minimize reading error. To ensure that systematic offsets between scales did not corrupt the data, we adopted a strict zeroing procedure: before each measurement sequence, the meter was zeroed on the 10 mW scale with the beam blocked. We then verified that this zero reading was preserved when switching to other scales (e.g., 20 mW). Any measurement for which the zero drifted upon scale switching was discarded, and the zeroing process was repeated until the zero remained stable across all scales used in that trial.

The total power uncertainty is obtained by combining the reading and accuracy errors in quadrature:[2]

$$\delta P = \sqrt{(\delta P_{\text{read}})^2 + (\delta P_{\text{acc}})^2}. \quad (15)$$

3. Transmittance Uncertainty

The transmittance is defined as the ratio of the dye solution power to the paired water baseline power:

$$T = \frac{P_{\text{dye}}}{P_{\text{water}}}. \quad (16)$$

Because P_{dye} and P_{water} are measured on potentially different scales and both include random linearity errors, the uncertainties in the numerator and denominator do *not* cancel. We therefore apply the general error propagation formula:[2]

$$\delta T = \sqrt{\left(\frac{\partial T}{\partial P_{\text{dye}}} \delta P_{\text{dye}}\right)^2 + \left(\frac{\partial T}{\partial P_{\text{water}}} \delta P_{\text{water}}\right)^2}. \quad (17)$$

The partial derivatives are

$$\frac{\partial T}{\partial P_{\text{dye}}} = \frac{1}{P_{\text{water}}}, \quad (18)$$

$$\frac{\partial T}{\partial P_{\text{water}}} = -\frac{P_{\text{dye}}}{P_{\text{water}}^2}, \quad (19)$$

yielding

$$\delta T = \sqrt{\left(\frac{\delta P_{\text{dye}}}{P_{\text{water}}}\right)^2 + \left(\frac{P_{\text{dye}} \delta P_{\text{water}}}{P_{\text{water}}^2}\right)^2}. \quad (20)$$

4. Absorbance Uncertainty

To linearize the exponential attenuation relationship predicted by the Beer-Lambert Law, we define the absorbance as

$$A = -\ln(T). \quad (21)$$

Propagating uncertainty through the logarithm:

$$\delta A = \left|\frac{dA}{dT}\right| \delta T = \left|\frac{-1}{T}\right| \delta T = \frac{\delta T}{T}. \quad (22)$$

5. Transformation to Logarithmic Space

To test the functional form predicted by the Beer-Lambert Law, we analyze our data in logarithmic space. Since $A = \varepsilon LC$, taking the natural logarithm of both sides yields

$$\ln(A) = \ln(\varepsilon) + \ln(L) + \ln(C), \quad (23)$$

which is linear in $\ln(L)$ and $\ln(C)$. Plotting $\ln(A)$ versus $\ln(C)$ (at fixed L) or $\ln(A)$ versus $\ln(L)$ (at fixed C) should produce straight lines with slopes of unity if the Beer-Lambert Law holds.

However, transforming to logarithmic coordinates also transforms the uncertainties. If a quantity x has absolute uncertainty δx , then the uncertainty in $\ln(x)$ is given by error propagation:

$$\delta(\ln x) = \left|\frac{d(\ln x)}{dx}\right| \delta x = \frac{\delta x}{x}. \quad (24)$$

Applying this to our variables:

$$\delta(\ln C) = \frac{\delta C}{C}, \quad (25)$$

$$\delta(\ln A) = \frac{\delta A}{A}. \quad (26)$$

These transformed uncertainties are used as weights in the fitting procedures described below.

E. Parameter Fitting Methodology

1. Linearization via Logarithmic Plotting

To empirically determine whether our data obey the Beer-Lambert Law and to extract the extinction coefficient ε , we performed linear regression on our absorbance data in logarithmic space. For fixed path length L , we fit the model

$$\ln(A) = m \cdot \ln(C) + b, \quad (27)$$

and for fixed concentration C , we fit

$$\ln(A) = m \cdot \ln(L) + b. \quad (28)$$

If the Beer-Lambert Law holds exactly, both slopes m should equal unity.

2. Orthogonal Distance Regression for Absorbance vs. Concentration

For the fits of $\ln(A)$ versus $\ln(C)$ (Eq. 27), both the independent and dependent variables carry non-negligible uncertainties. As detailed in the previous subsection, the concentration C has a relative uncertainty ranging from 0% (for the reference stock solution) to approximately 2.4% (for the most dilute solution), which propagates to $\delta(\ln C) = \delta C/C$. Similarly, the absorbance A has uncertainty $\delta A = \delta T/T$, yielding $\delta(\ln A) = \delta A/A$ in logarithmic space.

Standard weighted least-squares routines, such as `scipy.optimize.curve_fit`, are insufficient for this analysis. These methods assume the independent variable is known exactly ($x_{\text{true},i} = x_i$) and minimize only the vertical residuals between the data and the fitted curve. This assumption is violated when x carries significant uncertainty, as ignoring the horizontal error component produces biased parameter estimates and underestimates the true uncertainties in the fitted parameters.

To properly account for uncertainties in both variables, we employed Orthogonal Distance Regression (ODR), as implemented in the `scipy.odr` package[5]. ODR is a more general regression technique that does not assume the measured independent variable values are exact. Instead, for each data point (x_i, y_i) with uncertainties $(\delta x_i, \delta y_i)$, ODR simultaneously determines:

1. The best-fit model parameters (slope m and intercept b).
2. A set of “corrected” or “idealized” points $(x_{\text{true},i}, y_{\text{true},i})$ that lie exactly on the fitted curve, where $y_{\text{true},i} = f(x_{\text{true},i}; m, b)$.

The algorithm achieves this by minimizing the weighted sum of squared orthogonal distances between

the measured points and their corresponding idealized points on the model curve. The objective function is

$$\chi^2 = \sum_{i=1}^N \left[\left(\frac{x_i - x_{\text{true},i}}{\delta x_i} \right)^2 + \left(\frac{y_i - y_{\text{true},i}}{\delta y_i} \right)^2 \right], \quad (29)$$

where $x_i - x_{\text{true},i}$ represents the horizontal residual and $y_i - y_{\text{true},i}$ represents the vertical residual. By minimizing Eq. (29), ODR determines the model that requires the smallest overall adjustment to both coordinates, weighted by their respective uncertainties. This yields statistically unbiased estimates of the model parameters and properly propagated uncertainties in those parameters.

For each fixed path length L , we performed an independent ODR fit of $\ln(A)$ versus $\ln(C)$ across the five measured concentrations.

3. Weighted Least Squares for Absorbance vs. Path Length

In contrast to the concentration measurements, the path lengths L are determined by precision-manufactured optical cells and carry negligible uncertainty compared to all other error sources. Consequently, for fits of $\ln(A)$ versus $\ln(L)$ (Eq. 28), the independent variable can be treated as exact, and the full machinery of ODR is unnecessary.

For these fits, we employed standard weighted least-squares regression via `scipy.optimize.curve_fit`[5]. This routine minimizes the weighted sum of squared vertical residuals:

$$\chi^2 = \sum_{i=1}^N \left(\frac{y_i - f(x_i; m, b)}{\delta y_i} \right)^2, \quad (30)$$

where only the uncertainty in the dependent variable $y = \ln(A)$ is considered. The weights $1/\delta y_i^2$ ensure that data points with smaller uncertainties contribute more strongly to the fit.

For each fixed concentration C , we performed an independent weighted least-squares fit of $\ln(A)$ versus $\ln(L)$ across the six measured path lengths.

III. RESULTS AND ANALYSIS

A. Transmittance Measurements

We begin by examining the transmittance data as a function of both path length and concentration.

Examining Figures 3 and 4, we observe that the transmittance does not decrease linearly with either path length or concentration. Instead, both relationships exhibit a curved decay that one could guess to be characteristic of exponential functions. The data suggest functional forms approximately following $T \propto e^{-\alpha L}$ for fixed concentration and $T \propto e^{-\beta C}$ for fixed path length.

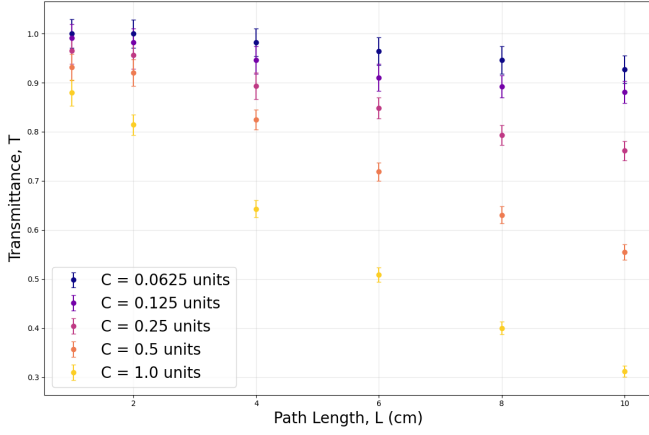


FIG. 3. Transmittance T versus path length L (linear scales) for five dye concentrations. Each concentration exhibits exponential decay in transmittance as path length increases, indicating absorption increases as more dye solution is present along the path of the laser, as expected.

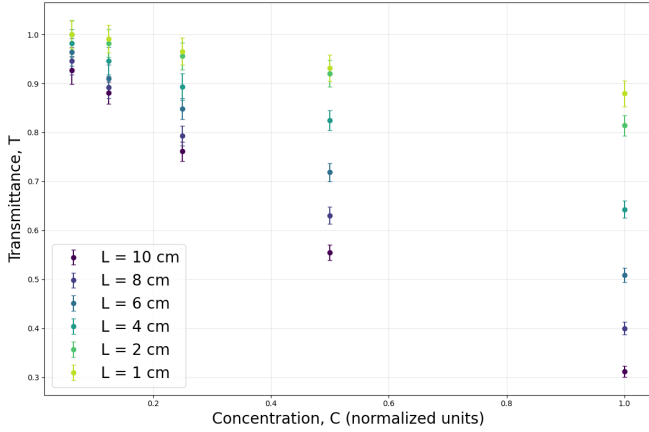


FIG. 4. Transmittance T versus concentration C (linear scales) for six path lengths. Each path length shows exponential decay in transmittance with increasing concentration, indicating that absorption strengthens with the number density of dye molecules.

Such exponential decay motivates a logarithmic transformation. If $T = e^{-\alpha L}$, then taking the natural logarithm yields $\ln(T) = -\alpha L$, or equivalently, $-\ln(T) = \alpha L$ —a linear relationship. Recall from the Introduction that we defined the absorbance precisely to linearize this exponential decay: $A = -\ln(T)$. The Beer-Lambert Law predicts that absorbance should be *linearly* proportional to both path length and concentration: $A = \varepsilon LC$. We now test this prediction by plotting absorbance in logarithmic coordinates.

B. Absorbance versus Concentration

To investigate the dependence of absorbance on concentration, we plot A versus C on log-log axes for each of the six path lengths. According to the Beer-Lambert Law, $A = \varepsilon LC$, which implies $\ln(A) = \ln(\varepsilon L) + \ln(C)$. Thus, a log-log plot of A versus C (at fixed L) should yield a straight line with slope $m = 1$.

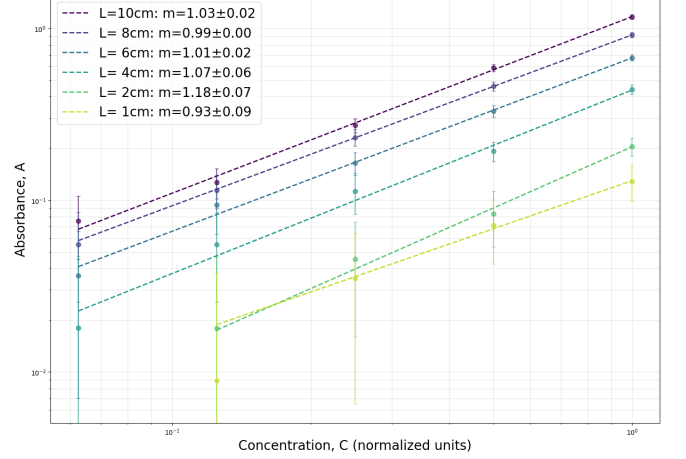


FIG. 5. Absorbance A versus concentration C on log-log axes for six path lengths. Error bars include propagated uncertainties from power and concentration measurements. Dashed lines represent Orthogonal Distance Regression (ODR) fits of $\ln(A) = m \cdot \ln(C) + b$ for each path length, accounting for uncertainties in both variables. Fitted slopes and uncertainties are displayed in the legend.

The fitted slopes obtained from ODR for each path length are summarized in Table I.

TABLE I. Fitted slopes m from log-log plots of absorbance versus concentration. Each fit was performed independently for a fixed path length using Orthogonal Distance Regression.

Path Length L (cm)	Slope m	Uncertainty σ_m	Weight
10	1.028	0.024	1736
8	0.995	0.004	62500
6	1.010	0.021	2268
4	1.069	0.063	252
2	1.182	0.074	183
1	0.928	0.086	135
			$\sum w_i$ 67074

To rigorously assess whether these slopes are consistent with the theoretical prediction $m = 1$, we calculate the weighted mean slope using inverse-variance weighting:[7]

$$\bar{m}_{\text{weighted}} = \frac{\sum_{i=1}^N w_i m_i}{\sum_{i=1}^N w_i}, \quad (31)$$

where the weights are $w_i = 1/\sigma_{m_i}^2$. The uncertainty in

the weighted mean is

$$\sigma_{\bar{m}} = \sqrt{\frac{1}{\sum_{i=1}^N w_i}}. \quad (32)$$

Applying these formulas to the data in Table I, we obtain:

$$\bar{m}_{\text{weighted}} = 1.003 \pm 0.004 [\text{U}]^{-1}. \quad (33)$$

To quantify the agreement with theory, we calculate the deviation in units of standard deviations:

$$z_m = \frac{m_{\text{exp}} - m_{\text{theory}}}{\sigma_{m_{\text{exp}}}} = \frac{1.003 - 1.000}{0.004} = 0.75 \sigma. \quad (34)$$

This confirms that the measured slope is consistent with the Beer-Lambert prediction to well within one standard deviation. The consistency across six independent path lengths—spanning from 1 to 10 cm—provides strong evidence that absorbance scales linearly with concentration in logarithmic space, as predicted by the Beer-Lambert Law.

C. Absorbance versus Path Length

Next, we examine the dependence of absorbance on path length by plotting A versus L on log-log axes for each of the five concentrations. A log-log plot of A versus L (at fixed C) should also yield a straight line with slope $m = 1$, according to: $\ln(A) = \ln(\varepsilon C) + \ln(L)$.

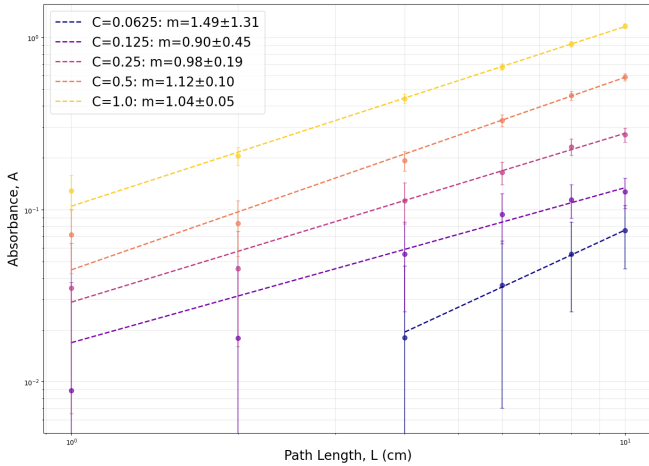


FIG. 6. Absorbance A versus path length L on log-log axes for five concentrations. Dashed lines represent weighted least-squares fits of $\ln(A) = m \cdot \ln(L) + b$ using `scipy.optimize.curve_fit`, as path length has negligible uncertainty.

The fitted slopes obtained from weighted least-squares regression are summarized in Table II. You may notice that as the concentration of the solution goes down, the uncertainties go up—a consequence of the power meter’s

TABLE II. Fitted slopes m from log-log plots of absorbance versus path length. Each fit was performed independently for a fixed concentration using weighted least-squares regression.

Concentration C (units)	Slope m	Uncertainty σ_m	Weight
0.0625	1.490	1.312	0.58
0.1250	0.901	0.454	4.85
0.2500	0.982	0.193	26.8
0.5000	1.119	0.098	104
1.0000	1.044	0.049	417
$\sum w_i$			553

readings between P_{water} and P_{dye} being similar, causing small values of T , and even smaller values of absorbance.

Calculating the weighted mean slope:

$$\bar{m}_{\text{weighted}} = 1.053 \pm 0.043 [\text{cm}]^{-1}. \quad (35)$$

The deviation from the theoretical prediction in units of standard deviations is:

$$z_m = \frac{m_{\text{exp}} - m_{\text{theory}}}{\sigma_{m_{\text{exp}}}} = \frac{1.053 - 1.000}{0.043} = 1.23 \sigma. \quad (36)$$

This result is consistent with the Beer-Lambert prediction of $m = 1$ to within approximately one standard deviation. The agreement is less precise than for the concentration-dependence analysis, primarily due to the larger uncertainties in the slopes for the lower concentrations (where absorbance values are small and relative errors are amplified). Nevertheless, the data across five independent concentrations—spanning a factor of 16 in dye concentration—confirm that absorbance scales linearly with path length in logarithmic space.

D. Intercept Analysis: Verification of Zero Absorbance at Limiting Cases

The Beer-Lambert Law implies that absorbance should vanish as either the concentration or the path length approaches zero. In logarithmic coordinates, the intercept b in the fit $\ln(A) = m \cdot \ln(x) + b$ does not directly correspond to $A = 0$ at $x = 0$, but we can still verify the physical expectation by examining whether the intercepts are consistent with the functional form $A = \varepsilon LC$ (which predicts $A \rightarrow 0$ as $L \rightarrow 0$ or $C \rightarrow 0$).

1. Intercepts from Absorbance versus Concentration Fits

The intercepts b obtained from the ODR fits of $\ln(A)$ versus $\ln(C)$ for each path length are listed in Table III.

All intercepts are consistent with zero to within one standard deviation. The intercept closest to zero ($b = 0.0004 \pm 0.0186$ for $L = 8$ cm) deviates by only 0.02σ , essentially indistinguishable from zero. The largest deviation ($b = -0.0128 \pm 0.0193$ for $L = 10$ cm) is 0.66σ , well

TABLE III. Fitted intercepts b from log-log plots of absorbance versus concentration.

Path Length L (cm)	Intercept b	Uncertainty σ_b	$ b/\sigma_b $
10	-0.0128	0.0193	0.66
8	0.0004	0.0186	0.02
6	-0.0018	0.0192	0.09
4	-0.0084	0.0196	0.43
2	-0.0127	0.0194	0.65
1	-0.0043	0.0196	0.22

within the expected statistical fluctuations. This consistency supports the expectation that no spurious offset or systematic error is present in the absorbance measurements across varying concentrations.

2. Intercepts from Absorbance versus Path Length Fits

Similarly, the intercepts from the weighted least-squares fits of $\ln(A)$ versus $\ln(L)$ for each concentration are shown in Table IV.

TABLE IV. Fitted intercepts b from log-log plots of absorbance versus path length.

Concentration C (units)	Intercept b	Uncertainty σ_b	$ b/\sigma_b $
0.0625	-0.0142	0.0226	0.63
0.1250	-0.0030	0.0223	0.13
0.2500	0.0003	0.0223	0.01
0.5000	-0.0239	0.0222	1.08
1.0000	-0.0145	0.0221	0.66

Once again, all intercepts are consistent with zero to within approximately one standard deviation. The intercept for $C = 0.25$ units ($b = 0.0003 \pm 0.0223$) is nearly perfectly zero (0.01σ), while the largest deviation ($b = -0.0239 \pm 0.0222$ for $C = 0.5$ units) is 1.08σ , just slightly above one standard deviation but still well within expected statistical scatter. These results confirm that the absorbance measurements across varying path lengths exhibit no systematic offset, further validating the Beer-Lambert functional form.

E. Determination of the Absorptivity Constant

Having verified that $A \propto C$ and $A \propto L$ independently, we now combine these dependencies to extract the proportionality constant—the absorptivity constant ε —from the full Beer-Lambert relationship $A = \varepsilon LC$.

A linear fit of A versus $L \cdot C$ yields:

$$\varepsilon = 0.1154 \pm 0.0025 \text{ [U}\cdot\text{cm}]^{-1}. \quad (37)$$

This value represents the absorptivity constant of the blue dye solution at the laser wavelength used in this

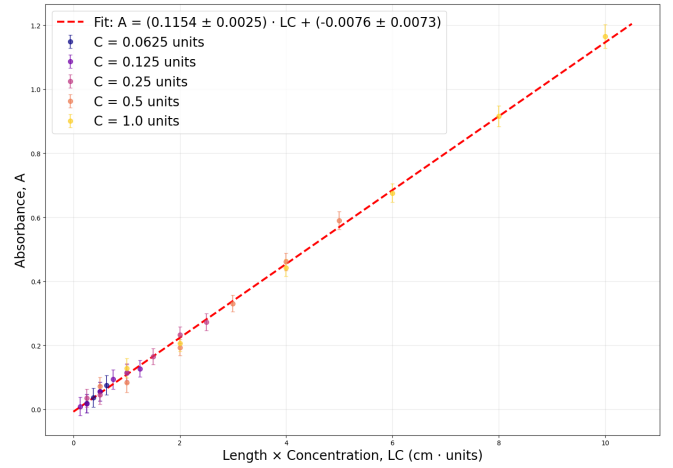


FIG. 7. Absorbance A versus the product of path length and concentration, $L \cdot C$ (linear scales). The dashed line represents a weighted least-squares fit of $A = \varepsilon(L \cdot C) + b$, yielding the absorptivity constant ε . Data points from all path lengths and concentrations are combined in a single linear regression.

experiment. The small relative uncertainty ($\sim 2.2\%$) reflects the precision of our measurement technique and error propagation. The linear relationship between A and $L \cdot C$ (Figure 7) visually confirms that the Beer-Lambert Law holds over the full range of our experimental parameters, with all data collapsing onto a single straight line as predicted.

F. Summary of Results

Table V provides a comprehensive summary of all key results obtained in this experiment.

TABLE V. Summary of experimental results testing the Beer-Lambert Law.

Quantity	Measured Value	Expected
<i>Slopes from $\ln(A)$ vs $\ln(C)$ fits (ODR)</i>		
Weighted mean slope	$1.003 \pm 0.004 \text{ [U]}^{-1}$	1.000 [U]^{-1}
Deviation from theory	0.75σ	—
Number of measurements	6 (at fixed L)	—
<i>Slopes from $\ln(A)$ vs $\ln(L)$ fits (curve_fit)</i>		
Weighted mean slope	$1.053 \pm 0.043 \text{ [cm]}^{-1}$	1.000 [cm]^{-1}
Deviation from theory	1.23σ	—
Number of measurements	5 (at fixed C)	—
<i>Intercepts from $\ln(A)$ vs $\ln(C)$ fits</i>		
Maximum deviation	0.66σ	0
All within	$< 1\sigma$	—
<i>Intercepts from $\ln(A)$ vs $\ln(L)$ fits</i>		
Maximum deviation	1.08σ	0
All within	$< 1.1\sigma$	—
<i>Absorptivity constant from A vs LC fit</i>		
ε	$0.1154 \pm 0.0025 \text{ [U}\cdot\text{cm}]^{-1}$	—
Relative uncertainty	2.2%	—

All experimental results are in excellent agreement

with the predictions of the Beer-Lambert Law. The weighted mean slopes from both sets of log-log fits are consistent with unity to better than 1.5σ , and all intercepts are consistent with zero to within approximately 1σ . These results validate the fundamental relationship $A = \varepsilon LC$ over path lengths ranging from 1 to 10 cm and concentrations spanning a factor of 16.

IV. CONCLUSION

This experiment successfully validated the Beer-Lambert Law, $A = \varepsilon LC$, over a wide range of experimental parameters. By measuring optical attenuation through blue dye solutions across path lengths from 1 to 10 cm and concentrations spanning a factor of 16, we confirmed that absorbance scales linearly with both variables. The weighted mean slopes from log-log fits of absorbance versus concentration and absorbance versus path length were $1.003 \pm 0.004 [\text{U}]^{-1}$ and $1.053 \pm 0.043 [\text{cm}]^{-1}$, respectively—both consistent with the theoretical prediction of unity to within 1.5σ . All fitted intercepts were consistent with zero, confirming that absorbance vanishes as either concentration or path length approaches zero. The extracted absorptivity constant, $\varepsilon = 0.1154 \pm 0.0025 [\text{U}\cdot\text{cm}]^{-1}$, was determined with a relative uncertainty of only 2.2%.

The success of our pairwise measurement approach—measuring P_{water} and P_{dye} sequentially in the same cell—was critical to achieving high precision. By forming the ratio $T = P_{\text{dye}}/P_{\text{water}}$, we effectively canceled the reflection losses at the air-glass and glass-liquid interfaces, isolating the absorption due solely to the dye molecules. This technique, combined with rigorous error propagation and the use of Orthogonal Distance Regression to account for uncertainties in both concentration and absorbance, ensured that our results are statistically robust.

A. Future Improvements and Extensions

Several experimental improvements could further enhance the precision of these measurements. First, using a higher initial stock concentration for the "1 unit" reference solution would allow us to explore a wider range of dilutions. At very low concentrations (e.g., $C = 0.0625$ units), the difference in transmitted power between water and dye solution becomes small, amplifying relative uncertainties. A more concentrated stock solution would increase this difference, improving signal-to-noise ratios and enabling measurements at even lower concentrations—potentially spanning factors of 32 or more rather than the factor of 16 achieved here.

Second, replacing the analog power meter with a digital oscilloscope would eliminate reading errors and reduce instrumental noise. The analog meter used in this experiment, while adequate, introduced uncertainties from

both needle positioning (± 1.0 unit) and inherent linearity limitations ($\sim 1\%$). A modern digital system could reduce these errors by an order of magnitude.

Finally, measuring the transmission coefficients T_1 (air-glass) and T_2 (glass-liquid) directly would provide a more complete characterization of the optical system. While our pairwise measurement approach successfully cancels these losses, independent knowledge of T_1 and T_2 would verify that the refractive index of the dilute dye solutions is indeed indistinguishable from that of pure water, which was an assumption we made in our experiment.

Beyond refining the current methodology, several interesting other experiments could be conducted. Testing the Beer-Lambert Law at very high concentrations could get us into the regime where molecular interactions or multiple scattering effects cause deviations from linearity. Using optical cells with longer path lengths (e.g., 20 or 50 cm) would extend the dynamic range and test the law over an even broader parameter space. Finally, exploring different dyes or mixtures would demonstrate the generality of the Beer-Lambert relationship across diverse systems.

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We would also like to acknowledge the use of Large Language Models (LLMs), specifically Google Gemini[8] and Claude AI[9]. They were helpful in the brainstorming process, in answering formatting questions for writing up the report in LaTeX and for assistance in making my plots look scientifically acceptable. It's important to note that these language models did not serve as a substitute for my own thinking and analysis; they were used as brainstorming partners and formatting aids.

Appendix A: Preliminary Methodology and Comparison

In an initial attempt to measure the Beer-Lambert Law, we employed a less rigorous experimental protocol that, while still yielding reasonable results, highlighted the importance of careful procedural design. This appendix documents the preliminary methodology, presents its results, and explains why a revised approach was ultimately adopted.

1. Description of Preliminary Methodology

The preliminary experiment differed from the final methodology in several critical ways:

Temporal separation of measurements: Water baseline measurements for all six cell lengths were conducted first, followed by dye solution measurements significantly later (on the order of tens of minutes to hours). This introduced potential time-dependent drift in the laser output power and photodetector sensitivity. Diode lasers are known to exhibit slow thermal drift as the junction temperature equilibrates, and ambient temperature fluctuations can cause the sensor response to vary over time.

Inconsistent zeroing protocol: The power meter was not zeroed on the same full-scale setting for both water and dye measurements. In some cases, the water baseline was measured on the 20 mW scale while the dye solution was measured on the 10 mW scale, without verifying that the zero reading was preserved across scale changes. This failure to maintain a consistent zero reference meant that any systematic offset between scales could propagate into the transmittance ratio.

Insufficient cleaning between trials: Optical cells were not cleaned as thoroughly between concentration changes, potentially leading to cross-contamination or residual dye film on the glass surfaces. Additionally, the measurement procedure varied slightly from trial to trial, reducing reproducibility.

Low stock concentration: The "1 unit" reference concentration was defined as a 80:20 mixture of stock dye with water. This choice reduced the dynamic range of the experiment; at the lowest concentration ($C = 0.0625$ units), the dye solution was nearly indistinguishable from pure water, yielding absorbance values close to zero with large relative uncertainties.

Despite these procedural shortcomings, the preliminary data still exhibited the qualitative features predicted by the Beer-Lambert Law. However, the quantitative agreement was less precise, and the results that were accurate spanned less path lengths and less concentrations.

2. Results from Preliminary Methodology

a. Absorptivity Constant

A linear fit of absorbance versus $L \cdot C$ for the preliminary dataset yielded:

$$\varepsilon_{\text{preliminary}} = 0.0744 \pm 0.0023 \text{ [U}\cdot\text{cm}]^{-1}, \quad (\text{A1})$$

The absorptivity constant used in the new experiment was $\varepsilon_{\text{improved}} = 0.1154 \pm 0.0025 \text{ [U}\cdot\text{cm}]^{-1}$. The improved methodology used a stock solution with approximately 55% higher dye concentration, which allowed us to get more accurate results for smaller concentrations.

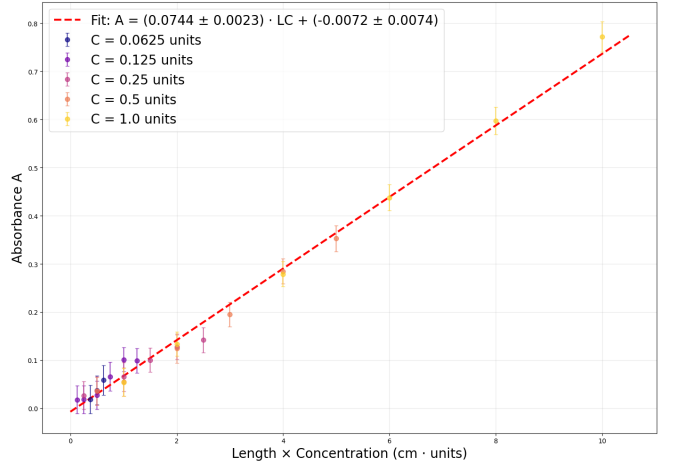


FIG. 8. Absorbance A versus $L \cdot C$ (linear scales) for the preliminary methodology. The dashed line represents a weighted least-squares fit yielding $\varepsilon = 0.0744 \pm 0.0023 \text{ [U}\cdot\text{cm}]^{-1}$. The lower slope compared to Figure 7 reflects the use of a more dilute stock solution.

b. Log-Log Slopes: Absorbance versus Path Length

Table VI summarizes the fitted slopes from log-log plots of absorbance versus path length for the preliminary dataset.

TABLE VI. Fitted slopes m from log-log plots of absorbance versus path length (preliminary methodology). Each fit was performed independently for a fixed concentration using weighted least-squares regression.

Concentration C (units)	Slope m	Uncertainty σ_m	Weight
0.0625	2.167	2.838	0.124
0.1250	0.917	0.505	3.921
0.2500	0.789	0.296	11.413
0.5000	1.115	0.171	34.199
1.0000	1.106	0.076	173.130
			$\sum w_i$ 222.787

Calculating the weighted mean slope using all five concentrations:

$$\bar{m}_{\text{weighted}} = 1.0884 \pm 0.0670 \text{ [cm]}^{-1}. \quad (\text{A2})$$

The deviation from the theoretical prediction is:

$$z_m = \frac{1.0884 - 1.000}{0.0670} = 1.32 \sigma. \quad (\text{A3})$$

This result is consistent with the Beer-Lambert prediction to within approximately one and a half standard deviations. However, the data at the lowest concentration ($C = 0.0625$ units) shows a slope of $m = 2.167 \pm 2.838 \text{ [cm]}^{-1}$, deviating by more than 4σ from unity and carrying enormous uncertainty. This is a direct consequence of the low stock concentration: at $C = 0.0625$ units, the absorbance values were so small that the relative uncertainties became unmanageably large, rendering

the fit unreliable. Excluding this outlier, the weighted mean becomes $\bar{m}_{\text{weighted}} = 1.058 \pm 0.065 \text{ [cm]}^{-1}$, with a deviation of 0.89σ from theory—a modest improvement in agreement, though the higher-concentration data already dominate the weighted average.

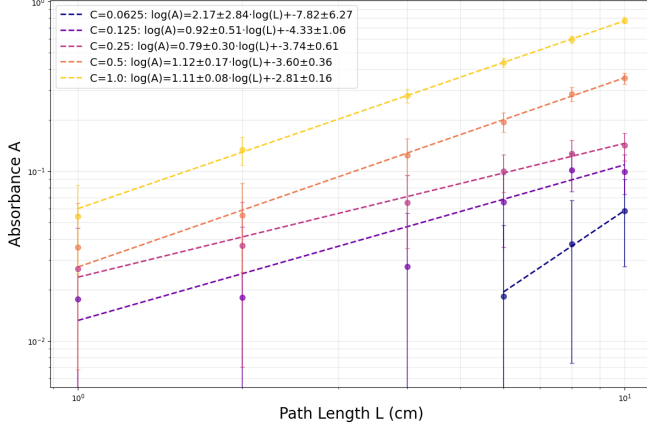


FIG. 9. Absorbance A versus path length L (log-log scales) for the preliminary methodology. The scatter and large uncertainties at low concentration reflect the reduced dynamic range of the dilute stock solution.

c. Log-Log Slopes: Absorbance versus Concentration

TABLE VII. Fitted slopes m from log-log plots of absorbance versus concentration (preliminary methodology). Each fit was performed independently for a fixed path length using Orthogonal Distance Regression.

Path Length L (cm)	Slope m	Uncertainty σ_m	Weight
10	1.082	0.067	223
8	0.987	0.073	188
6	1.054	0.062	260
4	1.101	0.039	658
2	1.026	0.110	83
1	0.531	0.027	1372
			$\sum w_i$ 2784

Table VII summarizes the fitted slopes from log-log plots of absorbance versus concentration.

The slope at $L = 1$ cm is a clear outlier, deviating from unity by more than 17σ . This low value likely reflects systematic errors specific to the shortest path length measurement in the preliminary protocol. Calculating the weighted mean with all six data points:

$$\bar{m}_{\text{weighted}} = 0.723 \pm 0.019 \text{ [U]}^{-1}. \quad (\text{A4})$$

The large weight assigned to the $L = 1$ cm measurement (due to its small uncertainty) pulls the weighted mean far from unity, yielding a 14.6σ deviation from theory. Excluding the outlier, $\bar{m}_{\text{weighted}} = 1.074 \pm 0.025 \text{ [U]}^{-1}$, with a deviation of 2.96σ . While still less precise than the improved methodology (0.75σ deviation), this demonstrates that the majority of the preliminary data were consistent with the Beer-Lambert Law when procedural errors were avoided.

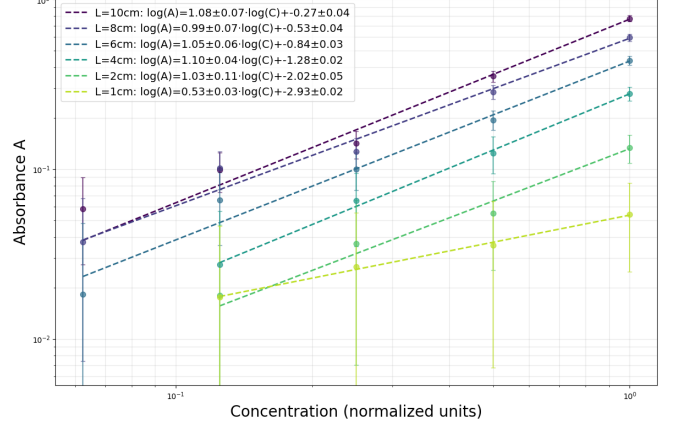


FIG. 10. Absorbance A versus concentration C (log-log scales) for the preliminary methodology. The anomalous behavior at $L = 1$ cm is visible as a significantly shallower slope compared to the other path lengths.

3. Justification for Revised Methodology

While the preliminary methodology produced results that qualitatively confirmed the Beer-Lambert Law, the quantitative agreement was shaky due to procedural inconsistencies. The temporal separation between water and dye measurements introduced potential drift, the lack of a rigorous zeroing protocol allowed systematic offsets to propagate, and the low stock concentration limited the dynamic range. Most importantly, the measurement protocol lacked repeatability: each trial was conducted slightly differently, making it difficult to isolate the source of discrepancies when they arose.

The revised methodology addressed all of these issues. By adopting a pairwise measurement approach—zeroing the meter, measuring water, immediately measuring dye in the same cell, then moving to the next cell—we eliminated temporal drift and ensured that both measurements experienced identical instrumental conditions. Verifying that the zero reading was preserved when switching scales removed systematic calibration offsets. Thorough cleaning between concentrations and strict adherence to a written protocol improved reproducibility. Finally, increasing the stock concentration to undiluted dye expanded the dynamic range, allowing reliable measurements down to $C = 0.0625$ units.

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